

Notes: Chodorow-Reich (QJE, 2014)
The Employment Effects of Credit Market Disruptions: Firm-level Evidence
from the 2008-9 Financial Crisis

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	3
3.1	Borrowers and loan relationships	3
3.2	Bank-health variables	4
3.3	Firm-level credit supply shock	4
3.4	Outcomes	4
4	Models and empirical specifications	5
4.1	Bank lending response	5
4.2	Borrower loan access	5
4.3	Employment effects	5
4.4	Heterogeneous effects by firm size	5
4.5	Placebo periods	5
5	Identification and empirical design	6
5.1	Core source of variation	6
5.2	Why pre-crisis relationships matter	6
5.3	Why “other-borrower lending” matters	6
5.4	Balance and unobserved borrower quality	6
5.5	Mechanism validation	6
5.6	External validity and limitations	6
6	Tables and figures	6
6.1	Table I: Banking Relationship Regressions; Table II: Sample Summary Statistics	6
6.2	Figure I: Stress in the Interbank Lending Market; Figure II: Aggregate New Lending from Top 43 Lenders	7
6.3	Table III: Determinants of Bank Lending; Table IV: Balancing of Covariates in the Sample; Table V: Testing for Unobserved Characteristics of Borrowers	8
6.4	Table VI: Effect of Bank Health on the Likelihood of Obtaining a Loan; Table VII: Effect of Bank Health on Interest Rate Spreads	8
6.5	Table VIII: Bond Market Access and Employment; Table IX: Effect of Lender Credit Supply on Employment	9
6.6	Table X: Employment Effects with Heterogeneous Size Effects; Figure III: Employment Losses by Firm Size	10
6.7	Table XI: Employment Effects Pre-Lehman and in the Placebo Period; Table XII: Placebo Periods; Table XIII: Total Effect of Credit Availability	11

7	Short summary	12
8	Conclusion	12
8.1	Core conclusion	12
8.2	Economic intuition	12
8.3	Identification lesson	13
8.4	Policy implication	13
8.5	Limitations	13
8.6	Takeaway	13

1 Big picture

- **Question.** Did the 2008-9 banking-sector disruption reduce employment at nonfinancial firms, and if so, how large was the effect?
- **Core identification idea.** Firms had pre-crisis lending syndicates. When those banks suffered different health shocks after Lehman, otherwise similar borrowers faced different credit-supply contractions.
- **Key treatment.** The paper constructs a firm-level predicted credit-supply shock using the change in lending by the banks in the firm's pre-crisis syndicate to *other* borrowers.
- **Main result.** Firms connected to healthier banks were more likely to obtain loans and experienced smaller employment losses.
- **Magnitude.** The estimates imply that credit supply explains a substantial share of employment losses at small and medium firms during the post-Lehman period.
- **Mechanism.** Banks exposed to the interbank freeze and asset-price collapse cut lending. Borrowers tied to those banks could not perfectly substitute to alternative financing, especially small and medium firms.
- **Scope.** The paper studies nonfinancial, non-FIRE U.S. firms with syndicated loan relationships before the crisis. The focus is bank credit supply, not demand for labor or aggregate demand alone.
- **Takeaway.** Bank health affects real employment through firm-level credit relationships; relationship banking can transmit financial-sector shocks to the labor market.

2 Incremental contribution of the paper

- **Firm-level identification of a macro-financial shock.** The paper connects the 2008 banking crisis to individual firms' employment outcomes using pre-crisis bank relationships, rather than relying only on aggregate time-series evidence.
- **Borrower-level credit supply measure.** It constructs a predicted credit-supply shock from lenders' changes in lending to *other* borrowers, reducing direct contamination from the firm's own demand shock.
- **Employment, not just credit quantities.** Earlier lending-channel papers often stop at loan volumes. This paper traces the shock through to employment growth.
- **Multiple validation layers.** The paper validates the bank-health channel using bank balance-sheet/market-exposure variables, loan likelihood, loan spreads, balance tests, placebo periods, and heterogeneous firm-size effects.
- **Quantification of crisis employment losses.** The paper translates micro estimates into a contribution to aggregate employment losses at small and medium firms.
- **Financial-frictions heterogeneity.** It shows that effects are largest for small and medium firms, consistent with limited substitution to public bond markets or alternative lenders.

Interpretation: The incremental contribution is the combination of a relationship-based credit-supply instrument and a real labor-market outcome. The paper shows that banking relationships can be a high-frequency transmission mechanism from financial-sector distress to firm employment.

3 Data and measurement

3.1 Borrowers and loan relationships

The sample consists of non-FIRE U.S. borrowers that obtained loans for working capital or general corporate purposes and had valid state, industry, and public/private status data. The main loan data come from

DealScan-LDB. A firm is connected to a pre-crisis syndicate if it borrowed from that bank group before the post-Lehman lending collapse.

Interpretation: The key empirical design uses pre-crisis relationships as predetermined links between firms and banks. These relationships determine which banks' later health shocks matter for the firm.

3.2 Bank-health variables

The paper studies several measures of bank health and exposure to crisis conditions:

- Lehman cosyndication exposure;
- ABX exposure;
- trading revenue/assets;
- 2007-8 real estate net charge-offs/assets;
- 2007 deposits/assets.

These measures help show that the banks cutting lending are those plausibly damaged by interbank stress, asset-price collapses, trading losses, or weak funding.

Interpretation: Bank health is not measured by a single accounting variable. The paper uses several pre-existing and crisis-relevant exposures to establish that bank balance-sheet distress predicts lending contraction.

3.3 Firm-level credit supply shock

For borrower i in pre-crisis syndicate s , the main exposure is the change in lending by the banks in that syndicate to *other* borrowers:

$$\widehat{\Delta L}_{i,s} = \text{change in lending by syndicate } s \text{ to borrowers other than } i.$$

The paper often uses the most liberal bank in the syndicate to measure credit availability, reflecting the idea that a borrower may need only one willing relationship lender to obtain credit.

Interpretation: Excluding the firm's own borrowing from the measure is crucial. It makes the measure closer to a supply shifter than to a joint supply-demand outcome for the same firm.

3.4 Outcomes

The main outcomes are:

- likelihood of obtaining a new loan or favorable loan modification;
- loan interest rate spreads;
- employment growth from 2008:3 to 2009:3;
- heterogeneous employment effects by firm size;
- placebo employment outcomes in pre-crisis periods.

Interpretation: The outcome sequence matches the hypothesized mechanism: bank health affects loan supply, loan supply affects borrowing, and borrowing constraints affect employment.

4 Models and empirical specifications

4.1 Bank lending response

A bank-level regression links lending changes to bank health:

$$\Delta L_b = \alpha + \beta Health_b + \Gamma' X_b + u_b.$$

Interpretation: This regression validates that the bank-health variables actually predict the contraction in new lending by the most active lenders.

4.2 Borrower loan access

For borrower i in syndicate s , loan access is modeled as:

$$P(Borrow_{i,s} = 1) = G\left(\pi_0 + \pi_1 \widetilde{\Delta L}_{i,s} + \gamma' X_i + \eta_{i,s}\right),$$

where $Borrow_{i,s}$ indicates a new loan or favorable modification during October 2008-June 2009.

Interpretation: A positive coefficient on $\widetilde{\Delta L}_{i,s}$ means that borrowers connected to lenders that continued lending to other firms were more likely to obtain credit themselves.

4.3 Employment effects

The baseline employment equation is:

$$g_{i,s,t-k,t}^y = \beta_0 + \beta_1 \widetilde{\Delta L}_{i,s} + \gamma' X_i + \varepsilon_{i,s,t-k,t},$$

where g^y is the symmetric growth rate of employment.

Interpretation: The coefficient β_1 maps the firm-level credit-supply shock into employment growth. The key identifying assumption is that, conditional on controls, $\widetilde{\Delta L}_{i,s}$ is not proxying for borrower demand.

4.4 Heterogeneous effects by firm size

The paper allows effects to differ by firm size:

$$g_i^y = \beta_0 + \beta_{1,small}(\widetilde{\Delta L}_i \times Small_i) + \beta_{1,med}(\widetilde{\Delta L}_i \times Medium_i) + \beta_{1,large}(\widetilde{\Delta L}_i \times Large_i) + \gamma' X_i + \varepsilon_i.$$

Interpretation: If credit frictions bind more for firms with fewer financing alternatives, effects should be strongest for small and medium firms. Table X tests this prediction.

4.5 Placebo periods

The placebo regressions apply the same credit-supply exposure to pre-crisis employment changes.

Interpretation: A failure to find placebo effects supports the interpretation that post-Lehman bank credit disruptions, not persistent borrower characteristics, drive the main results.

5 Identification and empirical design

5.1 Core source of variation

Identification comes from variation in the health of banks in a firm’s pre-crisis lending syndicate. Borrowers that depended on banks hit hard by the post-Lehman banking disruption faced weaker credit supply than borrowers tied to healthier banks.

5.2 Why pre-crisis relationships matter

Pre-crisis syndicates are formed before the employment shock and before the acute post-Lehman phase. This helps make bank exposure predetermined relative to post-Lehman employment outcomes.

5.3 Why “other-borrower lending” matters

A direct change in lending from bank b to borrower i would mix supply and demand. The paper therefore measures lender supply using the bank’s lending to other borrowers. This creates a leave-one-out credit-supply measure.

5.4 Balance and unobserved borrower quality

Table IV checks whether firms connected to constrained and unconstrained lenders differ on observables. Table V asks whether borrower characteristics not captured by controls predict bank supply exposure. These tests do not prove random assignment, but they reduce concerns that weak banks were simply matched with weak borrowers.

5.5 Mechanism validation

Tables III, VI, and VII validate the mechanism. Bank-health variables predict bank lending; predicted supply predicts the probability of obtaining a loan; and lower supply is associated with worse loan terms.

5.6 External validity and limitations

The analysis focuses on firms with syndicated loan relationships. These are larger than the average private firm, so the estimates are not direct effects for all firms. The aggregate exercise extrapolates to small and medium firms in the sample, not necessarily to the full economy without assumptions.

6 Tables and figures

6.1 Table I: Banking Relationship Regressions; Table II: Sample Summary Statistics

Read: Table I shows that firms’ pre-crisis banking relationships are persistent and explain future borrowing relationships. Table II describes the borrower and bank variables used in the main analysis.

Economic message: Relationship lending is central to the identification design. If pre-crisis relationships had no predictive power for later credit access, bank health would be a weak source of borrower variation.

Interpretation: Table I establishes that bank-firm links are not arbitrary. Table II shows that borrowers and banks are economically large and heterogeneous, providing enough variation to study the lending channel.

10

QUARTERLY JOURNAL OF ECONOMICS

TABLE I
BANKING RELATIONSHIP REGRESSIONS

	(1) Lender chosen as lead	(2) Lender chosen as lead	(3) Lender chosen as participant	(4) Lender chosen as participant
Explanatory variables				
Previous lead	0.71** (0.011)	0.67** (0.012)	0.022** (0.0040)	-0.023** (0.0045)
Previous participant	0.029** (0.0014)	0.020** (0.0015)	0.50** (0.011)	0.46** (0.011)
Previous lead × Public (Unrated)	-0.052** (0.016)	-0.043* (0.017)		
Previous lead × Public (Rated)	-0.058** (0.014)	-0.086** (0.016)		
Previous participant × Public (Unrated)			0.039* (0.018)	0.033+ (0.018)
Previous participant × Public (Rated)			0.012 (0.014)	-0.038* (0.015)
Lender FE	Yes	Yes	Yes	Yes
2-digit SIC × lender FE	No	Yes	No	Yes
State × lender FE	No	Yes	No	Yes
Year × lender FE	No	Yes	No	Yes
Public/private × lender FE	No	Yes	No	Yes
All in drawn quartile × lender FE	No	Yes	No	Yes
Sales quartile × lender FE	No	Yes	No	Yes
R ²	0.480	0.504	0.285	0.334
Borrower clusters	3,253	3,253	3,253	3,253
Observations	349,008	349,008	349,008	349,008

Table I: Banking relationship regressions.

The opening tables define the lending-relationship structure and the main borrower/bank variables.

TABLE II
SAMPLE SUMMARY STATISTICS

	N	Mean	Std. Dev.	p10	p50	p90
Panel A: Firm variables						
Loan size (millions of 2005 dollars)						
All lenders	4,791	287	530	23	119	693
Top 43 lenders	4,391	302	542	26	129	720
Merged Deutsche-LDB	2,940	305	544	27	131	703
Sales at close (millions of 2005 dollars)						
All lenders	3,954	1,836	4,059	53	433	4,661
Top 43 lenders	3,623	1,928	4,149	60	478	4,869
Merged Deutsche-LDB	1,721	2,094	4,310	68	551	4,835
Employment growth rate, 2008:3-2009:3	2,040	-0.09	0.23	-0.29	-0.06	0.08
2008 employment level	2,040	2,985	9,993	77	620	6,128
Panel B: Bank variables						
%A number of loans	43	-52.4	29.3	-87.4	-58.8	-7.5
Lehman cosyndication exposure (%)	42	1.15	1.20	0.34	0.72	1.91
ABX exposure	40	1.16	0.46	0.71	1.07	1.77
2007-8 trading revenue/assets (%)	42	-0.08	0.62	-0.72	0.01	0.39
2007-8 real estate net charge-offs/assets (%)	21	0.24	0.23	0.04	0.18	0.49
2007 deposits/assets (%)	43	42.2	25.4	3.0	47.4	68.2

Notes: The sample includes non-FRFR U.S. borrowers that obtained a loan for working capital or corporate purposes and with valid state, industry, and public/private status identifiers, excluding the top 15 by sales. Statistics for loan size and borrower size correspond to information as of the last loan obtained by each borrower prior to September 2008. Rows indicated by Top 43 lenders include only borrowers whose last previous loan included one of the most active 43 lenders as a lead arranger. Rows indicated by merged Deutsche-LDB and the employment statistics include only borrowers matched to the LDB. The variable %A loans equals the change in the estimated number of loans made by the bank between the periods October 2005 to June 2007 and October 2008 to June 2009. The variable Lehman cosyndication exposure equals the fraction of the bank's syndication

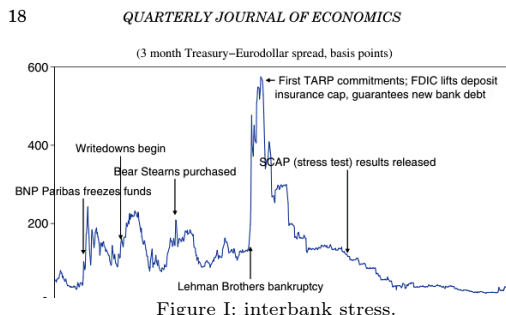
Table II: sample summary statistics.

6.2 Figure I: Stress in the Interbank Lending Market; Figure II: Aggregate New Lending from Top 43 Lenders

Read: Figure I shows the timing of interbank-market stress. Figure II shows the collapse in new lending by major lenders, especially around the post-Lehman period.

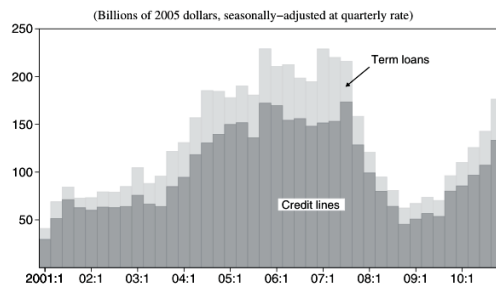
Economic message: The figures provide the aggregate crisis timeline. The empirical design then exploits how that aggregate shock differentially hit borrowers based on their bank relationships.

Interpretation: Figure I motivates the division between pre-Lehman and post-Lehman periods. Figure II shows that the credit crunch was quantitatively large in the syndicated-loan market.



The two figures motivate the timing and aggregate scale of the bank credit-supply shock.

EMPLOYMENT EFFECTS OF CREDIT MARKET DISRUPTIONS 29



6.3 Table III: Determinants of Bank Lending; Table IV: Balancing of Covariates in the Sample; Table V: Testing for Unobserved Characteristics of Borrowers

Read: Table III shows that crisis-relevant bank-health variables predict changes in bank lending. Table IV checks observable balance across lender-health quartiles. Table V tests whether borrower unobservables appear correlated with lender health exposure.

Economic message: These tables are the main credibility checks for the credit-supply shock. They show that bank health predicts lending and that borrower composition is not obviously driving the result.

Interpretation: The empirical design requires weak banks to reduce lending for bank-side reasons, not because their borrowers had worse demand. These tables directly address that concern.

26

QUARTERLY JOURNAL OF ECONOMICS

QUARTERLY JOURNAL OF ECONOMICS			
TABLE III DETERMINANTS OF BANK LENDING			
	(1)	(2)	(3)
Change in lending during the crisis			
Explanatory variables			
Lehman cosyndication exposure	-0.14** (0.049)		
ABX exposure		-0.11* (0.041)	
2007–8 trading revenues/assets			0.046 (0.040)
Real estate charge-offs flag			0.012 (0.050)
2007–8 real estate net charge-offs/assets			-0.092* (0.051)
2007 Bank Deposits/Assets			0.19** (0.059)
Joint test p -value	0.008	0.013	0.002
R^2	0.16	0.15	0.35
Observations	42	40	42

Notes. The dependent variable is the change in the annualized number of loans made by the bank between the periods October 2005 to June 2007 and October 2008 to June 2009, with each loan scaled by the importance of the lender in the loan portfolio as described in Section IV.C of the text. Observations weighted by amount of precrisis business. The explanatory variables have been normalized to have unit variances. *, **, and *** indicate significance at the 0.1, 0.05, and 0.01 levels, respectively.

Table III: bank lending determinants.

TABLE IV BALANCING OF COVARIATES IN THE SAMPLE					
Quantile of lender health					Mean: std. dev.
1	2	3	4		
Mean employment change in					
Borrower's industry, 2008:3–2009:3					
-0.081	-0.081	-0.085	-0.089	0.083	
-0.056	-0.056	-0.056	-0.056	0.009	
Share with bond market access					
0.455	0.540	0.458	0.236	0.494	
Share private, no bond market access					
0.418	0.311	0.363	0.525	0.492	
Share public, no bond market access					
0.127	0.129	0.179	0.239	0.374	
Mean all in draws spread					
266	155	156	199	133	
Median sales at close (\$2005 billions)					
0.966	0.857	0.701	0.285	4.146	
Mean year of last precrisis loan					
2005.83	2005.08	2006.03	2006.05	1.50	
Share with loan due during crisis					
0.193	0.188	0.183	0.205	0.394	

Notes. The table splits the sample into four quartiles based on the change in the annualized number of loans made by the borrower's last precrisis creditors between the periods October 2005 to June 2007 and October 2008 to June 2009. Employment change by borrower industry computed at the four-digit SIC level using six-digit NAICS employment levels from the Quarterly Census of Wage and Employment and a SIC NAICS concordance table available from the BLS. Employment change by borrower county computed by averaging the employment change in all counties in which a firm operates establishments using establishment employment shares as weights. The last column reports the standard deviation of the variable summarized in each row.

the top and bottom quartile operated in counties that had essentially identical average employment changes.²⁰

The fact that lenders identified as healthier did not lend before the crisis to firms in counties or industries that had systematically better crisis employment outcomes indicates that bank specialization by industry or geography cannot explain the borrower outcomes even in a bivariate ordinary least squares (OLS) context. To measure the balance of differences between

Table IV: balance.

			(1)	(2)
			Δ Log (lending in borrower-lender pair)	
Explanatory variables				
% Δ loans to other borrowers (ΔL_i)			1.05**	1.071**
			(0.33)	(0.32)
1-digit SIC, loan year FE			No	Yes
Bond market access/public/private FE			No	Yes
Additional Dealscan controls			No	Yes
Borrower FE			No	No
R^2			0.423	0.088
Borrowers			432	432
Banks			43	43
Observations			2,005	2,005

Notes. The sample contains only borrowers that signed a new loan between October 2008 and June 2009.

Table V: unobserved borrower test.

TABLE VI
THE EFFECT OF BANK HEALTH ON THE LIKELIHOOD OF OBTAINING A LOAN

	(1)	(2)	(3)	(4)	(5)	(6)
	Firm obtains a new loan or positive modification					
	Probit					
	$\Delta \tilde{L}_{i,s}$ instrumented using					
	Lehman exposure		ABX exposure	Bank statement items		
						All
Explanatory variables						
% Δ loans to other firms ($\Delta \tilde{L}_{i,s}$)	2.19** (0.79)	2.00** (0.53)	3.65** (1.28)	2.33* (1.12)	2.28** (0.64)	2.32** (0.63)
2-digit SIC, state, loan year FE	No	Yes	Yes	Yes	Yes	Yes
Bond access/public/private FE	No	Yes	Yes	Yes	Yes	Yes
Additional Dealscan controls	No	Yes	Yes	Yes	Yes	Yes
First stage F -statistic			14.0	8.2	18.2	19.8
J -statistic p -value						0.206
$E[\text{borrow}]$	0.134	0.134	0.134	0.134	0.134	0.134
$E[\text{borrow}:\Delta \tilde{L}_{p0} - \Delta \tilde{L}_{p10}]$	0.052	0.048	0.087	0.055	0.054	0.055
Lead lender 1 clusters	43	43	43	40	43	40
Lead lender 2 clusters	43	43	43	40	43	40
Observations	4,391	4,391	4,391	4,354	4,391	4,354

Notes. The dependent variable is an indicator for whether the borrower signed a new loan or received a favorable modification to an existing loan between October 2008 and June 2009. The variable $\Delta \tilde{L}_{i,s}$ equals the change in the annualized number of loans made by the bank between the periods October 2005 to June 2007 and October 2008 to June 2009, and has been normalized to have unit variance. The variable Lehman cosyndication exposure equals the fraction of the bank's syndication portfolio where Lehman Brothers had a lead role in the loan deal. The variable ABX exposure equals the loading of the bank's stock return on the ABX AAA 2006-H1 index between October 2007 and December 2007. The balance sheet and income statement items include the ratio of deposits to assets at the end of 2007, the ratio of trading revenue over 2007-8 to assets, the ratio of net real estate charge-offs over 2007-8 to assets, and an indicator for reporting real estate charge-offs. The last column includes all of the instruments. For each firm, the bank-level measures are averaged over the members of the firm's last precrisis loan syndicate, with weights given according to each bank's role. In columns (1) and (2) estimation is via probit, and the table reports marginal coefficients. In columns (3)-(6) $\Delta \tilde{L}_{i,s}$ is instrumented using the variable indicated in the column heading and estimation is via two-stage least squares. Borrower-level covariates are as of the last precrisis loan taken by each borrower. Additional Dealscan controls: multiple lead lenders indicator, loan due during crisis indicator, credit line indicator, log sales at close, all in drawn spread, credit line * all in drawn. Standard errors in parentheses and two-way clustered on the lead lenders in the borrower's last precrisis loan syndicate. *, **, and *** indicate significance at the 0.1, 0.05, and 0.01 levels, respectively.

in Section IV.D; and $\eta_{i,s}$ is an error term. The estimation uses the full Dealscan sample (i.e., including firms not matched to the LDB) and is done via probit. The table reports the marginal coef-

Table VI: loan likelihood.

The loan-access and loan-price tables validate the borrower-level financial channel.

6.5 Table VIII: Bond Market Access and Employment; Table IX: Effect of Lender Credit Supply on Employment

Read: Table VIII shows that firms without bond-market access have lower employment growth. Table IX estimates the main employment effect of lender credit supply.

Economic message: The employment effect is larger when alternative financing is limited. Table IX is the baseline real-effect regression.

Interpretation: Table VIII motivates why bank loans matter: firms with public debt access can substitute more easily. Table IX shows that predicted loan supply matters for employment after controls.

TABLE VII
THE EFFECT OF BANK HEALTH ON INTEREST RATE SPREADS

	(1)	(2)	(3)	(4)	(5)	(6)
	Change in interest rate spread					
	OLS					
	$\Delta \tilde{L}_{i,s}$ instrumented using					
	Lehman exposure		ABX exposure	Bank statement items		
						All
Explanatory variables						
% Δ loans to other firms ($\Delta \tilde{L}_{i,s}$)	-14.6** (5.26)	-12.2** (4.15)	-23.1* (11.2)	-20.0 (13.3)	-17.2* (7.63)	-17.6** (6.68)
1-digit SIC, loan year FE	No	Yes	Yes	Yes	Yes	Yes
Bond access/public/private FE	No	Yes	Yes	Yes	Yes	Yes
Additional Dealscan controls	No	Yes	Yes	Yes	Yes	Yes
First stage F -statistic			60.5	7.8	14.3	14.5
J -statistic p -value						0.967
$E[\Delta \text{Spread}]$	130.6	130.6	130.6	130.7	130.6	130.7
$E[\text{Spread}:\Delta \tilde{L}_{p0} - \Delta \tilde{L}_{p10}]$	-39.7	-33.0	-62.8	-54.3	-46.6	-47.7
Lead lender 1 clusters	34	34	34	32	34	32
Lead lender 2 clusters	30	30	30	28	30	28
Observations	350	350	350	346	350	346

Notes. The dependent variable is the interest spread, in basis points, charged to a firm on a loan starting between October 2008 and June 2009, less the interest spread charged to the same firm on its last loan of the same type (credit line or term loan) obtained prior to September 15, 2008. The regressions exclude loan pairs with an increase of >400 basis points. See the text for further details of the sample construction. The variable $\Delta \tilde{L}_{i,s}$ equals the change in the annualized number of loans made by the bank between the periods October 2005 to June 2007 and October 2008 to June 2009 and has been normalized to have unit variance. The variable Lehman cosyndication exposure equals the fraction of the bank's syndication portfolio where Lehman Brothers had a lead role in the loan deal. The variable ABX exposure equals the loading of the bank's stock return on the ABX AAA 2006-H1 index between October 2007 and December 2007. The balance sheet and income statement items include the ratio of deposits to assets at the end of 2007, the ratio of trading revenue over 2007-8 to assets, the ratio of net real estate charge-offs over 2007-8 to assets, and an indicator for reporting real estate charge-offs. For each firm, the bank-level measures are averaged over the members of the firm's last precrisis loan syndicate, with weights given according to each bank's role. Additional Dealscan controls: multiple lead lenders indicator, loan due during crisis indicator, credit line indicator, log sales at close, all in drawn spread, credit line * all in drawn. Standard errors in parentheses and two-way clustered on the lead lenders in the borrower's last precrisis loan syndicate. *, **, and *** indicate significance at the 0.1, 0.05, and 0.01 levels, respectively.

controls have minor effect on the point estimate. The instrumental variables specifications suggest even larger effects on borrow-

Table VII: loan interest rate spreads.

TABLE IX
THE EFFECT OF LENDER CREDIT SUPPLY ON EMPLOYMENT

	(1)	(2)	(3)	(4)	(5)	(6)
	Employment growth rate 2008:3-2009:3					
	OLS		$\Delta \tilde{L}_{i,t}$ instrumented using			
			Lehman exposure	ABX exposure	Bank statement items	All
Explanatory variables						
% Δ loans to other firms ($\Delta \tilde{L}_{i,t}$)	1.17* (0.58)	1.67** (0.61)	2.49* (1.00)	3.17* (1.35)	2.13* (0.88)	2.38** (0.77)
Lagged employment growth		0.0033 (0.019)	0.0039 (0.019)	0.0045 (0.019)	0.0036 (0.019)	0.0039 (0.019)
Emp. change in firm's county		0.89* (0.43)	0.85+ (0.46)	0.86+ (0.48)	0.87+ (0.45)	0.89+ (0.46)
2-digit SIC, state, loan year FE	No	Yes	Yes	Yes	Yes	Yes
Firm size bin FE	No	Yes	Yes	Yes	Yes	Yes
Firm age bin FE	No	Yes	Yes	Yes	Yes	Yes
Bond access/public/private FE	No	Yes	Yes	Yes	Yes	Yes
Additional Dealscan controls	No	Yes	Yes	Yes	Yes	Yes
First-stage F-statistic			15.5	8.5	18.5	23.1
J-statistic p-value						0.190
$E[g_t^*]$	-0.092	-0.092	-0.092	-0.093	-0.092	-0.093
$E[g_t^* \Delta \tilde{L}_{pnt} - \Delta \tilde{L}_{pnt}]$	0.027	0.039	0.058	0.074	0.050	0.055
Lead lender 1 clusters	43	43	43	40	43	40
Lead lender 2 clusters	43	43	43	40	43	40
Observations	2,040	2,040	2,040	2,015	2,040	2,015

Notes. The dependent variable is the symmetric growth rate g_t^* of employment. The variable $\Delta \tilde{L}_{i,t}$ equals the change in the annualized number of loans made by the bank between the periods October 2005 to June 2007 and October 2008 to June 2009 and has been normalized to have unit variance. The variable Lehman co-syndication exposure equals the fraction of the bank's syndication portfolio where Lehman Brothers had a lead role in the loan deal. The variable ABX exposure equals the loading of the bank's stock return on the ABX AAA 2006-H1 index between October 2007 and December 2007. The balance sheet and income statement items include the ratio of deposits to assets at the end of 2007, the ratio of trading revenue over 2007-8 to assets, the ratio of net real estate charge-offs over 2007-8 to assets, and an indicator for report real estate charge-offs. For each firm, the bank-level measures are averaged over the members of the firm's last precrisis loan syndicate, with weights given according to each bank's role. In columns (1) and (2) estimation is via OLS. In columns (3)-(6) $\Delta \tilde{L}_{i,t}$ is instrumented using the variable indicated in the column heading. Borrower-level covariates are as of the last precrisis loan taken by each borrower. Firms divided into size bin classes of 1-250, 250-999, and 1,000+, and age bins for birth in the 2000s, 1990s, or earlier. Eicker-White standard errors in parentheses. +, *, and ** indicate significance at the 0.1, 0.05, and 0.01 levels, respectively.

decline in the sample equaled 9.2%, and employment in the total
Table IX: credit supply and employment.

EMPLOYMENT EFFECTS OF CREDIT MARKET DISRUPTIONS 39

TABLE VIII
BOND MARKET ACCESS AND EMPLOYMENT

	(1)	(2)
	Employment growth rate 2008:3-2009:3	
Explanatory variables		
No bond market access	-2.65** (0.98)	-3.15** (1.10)
2-digit SIC and state FE	No	Yes
Firm size bin FE	No	Yes
Firm age bin FE	No	Yes
Lagged employment growth	No	Yes
County employment growth	No	Yes
R ²	0.003	0.172
Observations	2,040	2,040

Notes. The dependent variable is the symmetric growth rate g_t^* of employment. Firms that do not have access to the bond market do not have a credit rating from either Moody's or Standard and Poors, and have never issued public debt. Firms divided into size bin classes of 1-250, 250-999, and 1,000+, and age bins for birth in the 2000s, 1990s, or earlier. Eicker-White standard errors in parentheses. +, *, and ** indicate significance at the 0.1, 0.05, and 0.01 levels respectively.

Table VIII: bond-market access.

These tables connect financing access to employment outcomes.

6.6 Table X: Employment Effects with Heterogeneous Size Effects; Figure III: Employment Losses by Firm Size

Read: Table X allows the credit-supply effect to differ by firm size. Figure III shows that employment losses during the crisis were largest for smaller firms.

Economic message: The heterogeneous-effects result matches the financial-frictions story: small and medium firms are more bank-dependent and have fewer financing substitutes.

Interpretation: The large-firm estimates are small and imprecise, while small/medium firms show the relevant employment response. This is why the aggregate translation focuses on small and medium firms.

TABLE X
THE EFFECT OF LENDER CREDIT SUPPLY ON EMPLOYMENT WITH HETEROGENEOUS
TREATMENT EFFECTS

	(1)	(2)	(3)
	Employment growth rate 2008:3–2009:3		
Explanatory variables			
$\Delta \tilde{L}_{i,s}$ * Large	0.54 (0.97)		
$\Delta \tilde{L}_{i,s}$ * Medium	1.84+ (0.97)		
$\Delta \tilde{L}_{i,s}$ * Small	2.16** (0.79)		
$\Delta \tilde{L}_{i,s}$ * Bond market access		1.04 (1.00)	
$\Delta \tilde{L}_{i,s}$ * No access		2.01** (0.60)	
$\Delta \tilde{L}_{i,s}$ * Bond access & large			0.23 (1.15)
$\Delta \tilde{L}_{i,s}$ * Bond access & small/medium			1.47 (1.06)
$\Delta \tilde{L}_{i,s}$ * No access & large			0.79 (1.21)
$\Delta \tilde{L}_{i,s}$ * No access & small/medium			2.26** (0.58)
Lagged employment growth	Yes	Yes	Yes
Emp. change in firm's county	Yes	Yes	Yes
2-digit SIC, state, loan year FE	Yes	Yes	Yes
Firm size and age bin FE	Yes	Yes	Yes
Bond access/public/private FE	Yes	Yes	Yes
Additional Dealscan controls	Yes	Yes	Yes
Observations (Access & large)	483	483	483
Observations (Access & small/medium)	434	434	434
Observations (No access & large)	315	315	315
Observations (No access & small/medium)	808	808	808
Observations	2,040	2,040	2,040

Table X: heterogeneous size effects.

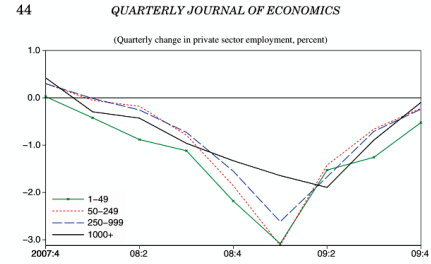
The size heterogeneity evidence links bank dependence to employment losses.

6.7 Table XI: Employment Effects Pre-Lehman and in the Placebo Period; Table XII: Placebo Periods; Table XIII: Total Effect of Credit Availability

Read: Tables XI and XII show that the main credit-supply-employment relation is concentrated in the post-Lehman crisis period and not in placebo periods. Table XIII translates the estimates into the fraction of sample employment losses attributable to credit availability.

Economic message: The placebo tests reduce the concern that lender exposure is simply proxying for persistent borrower trajectories. Table XIII turns the micro estimates into an aggregate employment-loss contribution.

Interpretation: The absence of comparable pre-crisis effects is critical. The effect appears when bank credit markets are disrupted, which is consistent with a credit-supply channel rather than time-invariant borrower selection.



The figure shows the percent change in employment by firm size class. The
Figure III: employment losses by firm size.

TABLE XI
THE EFFECT OF LENDER CREDIT SUPPLY ON EMPLOYMENT PRE-LEHMAN AND IN THE MEDIUM RUN

	(1)	(2)	(3)	(4)	(5)
	Employment growth rate				
	OLS				
	$\Delta \tilde{L}_{i,t}$ instrumented using				
	Lehman exposure	ABX exposure	Bank statement items	All	
Panel A: 2007:4-2008:3					
Explanatory variables					
% Δ loans to other firms ($\Delta \tilde{L}_{i,t}$)	0.55+	1.26			
	(0.31)	(0.81)			
Lagged employment growth	0.052**	0.053**			
	(0.015)	(0.015)			
Emp. change in firm's county	0.59+	0.52			
	(0.32)	(0.32)			
First-stage F -statistic		9.4			
Observations	1,895	1,872			
Panel B: 2008:3-2010:3					
Explanatory variables					
% Δ loans to other firms ($\Delta \tilde{L}_{i,t}$)	1.94**	3.40**	5.18**	2.14*	2.67**
	(0.63)	(1.26)	(1.94)	(1.00)	(0.90)
Lagged employment growth	0.049**	0.051**	0.052**	0.050**	0.050**
	(0.018)	(0.018)	(0.017)	(0.017)	(0.017)
Emp. change in firm's county	-0.17	-0.21	-0.25	-0.17	-0.19
	(0.49)	(0.52)	(0.52)	(0.50)	(0.50)
First-stage F -statistic		15.4	8.4	18.5	23.0
Observations	2,013	2,013	1,988	2,013	1,988
2-digit SIC, state, loan year FE	Yes	Yes	Yes	Yes	Yes
Firm size and age bin FE	Yes	Yes	Yes	Yes	Yes
Bond access/public/private FE	Yes	Yes	Yes	Yes	Yes
Additional Dealscan controls	Yes	Yes	Yes	Yes	Yes

Notes. The dependent variable is the symmetric growth rate g_t^e of employment. The variable $\Delta \tilde{L}_{i,t}$ equals the change in the annualized number of loans made by the bank between the periods December 2004 to August 2006 and December 2007 to August 2008 (Panel A), or between the periods October 2008 to June 2009 and October 2009 to June 2010 (Panel B) and has been normalized to have unit variance. Firms divided into size bin classes of 1-20, 20-500, and 1,000+, and age bins for both in the 2000s, 1990s, or earlier. Additional Dealscan controls: multiple lead lenders indicator, loan due during crisis indicator, credit line indicator, log sales at close, all in drawn spread, credit line + all in drawn. Standard errors in parentheses and two-way clustered on the lead lenders in the borrower's last precrisis loan syndicate. +, *, and ** indicate significance at the 0.1, 0.05, and 0.01 levels, respectively.

Table XI: pre-Lehman/placebo.

The final tables support timing and quantify aggregate implications for the sample.

7 Short summary

- The paper studies whether bank health during the 2008-9 crisis affected firm employment.
- It uses pre-crisis lending syndicates to assign firms to bank credit-supply shocks.
- Bank health predicts reduced lending during the post-Lehman period.
- Borrowers tied to healthier lenders are more likely to obtain loans or favorable modifications.
- Credit supply affects employment, especially at small and medium firms.
- Firms without bond-market access are more exposed to bank credit disruptions.
- Placebo periods do not show the same employment effects.
- The paper estimates that credit availability accounts for a sizable share of sample employment losses at small and medium firms.

8 Conclusion

8.1 Core conclusion

Chodorow-Reich shows that bank credit supply mattered for firm employment during the 2008-9 financial crisis. Firms connected to healthier pre-crisis lenders obtained more credit and suffered smaller employment losses.

8.2 Economic intuition

Relationship lenders are not perfectly substitutable in a crisis. When a firm's existing banks become distressed, the firm may not immediately replace them with new lenders or bond-market financing. This credit disruption affects employment because working capital and loan renewals support ongoing operations.

TABLE XII
THE EFFECT OF LENDER CREDIT SUPPLY ON EMPLOYMENT IN TWO PLACEBO PERIODS

	(1)	(2)	(3)	(4)	(5)
	Employment growth rate				
	OLS				
	$\Delta \tilde{L}_{i,t}$ instrumented using				
	Lehman exposure	ABX exposure	Bank statement items	All	
Panel A: 2005:2-2007:2					
Explanatory variables					
% Δ loans to other firms ($\Delta \tilde{L}_{i,t}$)	-0.19	-0.67	-1.57	1.63	0.92
	(0.74)	(1.63)	(1.72)	(1.24)	(1.15)
Lagged employment growth	0.028+	0.027+	0.028+	0.028+	0.028+
	(0.014)	(0.014)	(0.014)	(0.015)	(0.015)
Emp. change in firm's county	0.80	0.80	0.78	0.79	0.77
	(0.49)	(0.49)	(0.50)	(0.48)	(0.49)
First-stage F -statistic		15.6	8.8	18.9	23.8
Observations	1,879	1,879	1,854	1,879	1,854
Panel B: 2001:3-2002:3					
Explanatory variables					
% Δ loans to other firms ($\Delta \tilde{L}_{i,t}$)	-0.80	-0.74	1.30	-0.93	-0.72
	(0.59)	(1.44)	(1.89)	(0.93)	(0.85)
Lagged employment growth	0.024	0.024	0.024	0.024	0.024
	(0.020)	(0.020)	(0.020)	(0.020)	(0.020)
Emp. change in firm's county	1.53**	1.53**	1.62**	1.53**	1.59**
	(0.51)	(0.50)	(0.51)	(0.51)	(0.50)
First-stage F -statistic		16.5	7.7	17.8	26.3
Observations	1,675	1,675	1,653	1,675	1,653
2-digit SIC, state, loan year FE	Yes	Yes	Yes	Yes	Yes
Firm size and age bin FE	Yes	Yes	Yes	Yes	Yes
Bond access/public/private FE	Yes	Yes	Yes	Yes	Yes
Additional Dealscan controls	Yes	Yes	Yes	Yes	Yes

Table XII: placebo periods.

	2008:3-2009:3 (%)
Total employment decline	7.0
Share of losses due to credit availability, $\tau \leq 90$	34.4
Share of losses due to credit availability, $\tau \leq 95$	47.3

Notes. The table reports the fraction of employment losses due to credit availability at small and medium firms, as identified in the text. τ refers to the percentile of the banking syndicate identified in the next three syndicates.

Table XIII: total effect.

8.3 Identification lesson

The key design insight is to use lender health and pre-crisis relationships to generate firm-level variation in credit supply. The leave-one-out syndicate measure helps separate lender supply shocks from borrower demand shocks.

8.4 Policy implication

Bank capital, liquidity support, and lender stabilization can have real employment effects. Policies aimed at stabilizing banks may protect employment at relationship-dependent firms, especially smaller firms without bond-market access.

8.5 Limitations

The sample consists of firms with syndicated loans and therefore does not cover all small businesses. The credit-supply measure depends on pre-crisis syndicate structure, and bank health may still correlate with unobserved borrower characteristics despite balance and placebo tests.

8.6 Takeaway

The paper provides firm-level evidence that financial-sector disruptions can transmit to the real economy through relationship lending. Banking shocks do not only affect loan quantities; they affect jobs.